

Table 2: BOOOOKSCORE for summaries generated under different configurations; higher scores indicate better coherence. We additionally report the *average summary length* in tokens based on tiktoken (<https://github.com/openai/tiktoken>) tokenizer, the *percentage of novel trigrams* compared to the source, and *percentage of repeated trigrams* in the summary.

Model	Chunk size	BOOOOKSCORE	Avg. length	% novel 3-grams	% rep. 3-grams
<i>Summaries generated via hierarchical merging</i>					
GPT-4	2048	89.1	778.6	82.4	4.2
GPT-3.5-Turbo	2048	84.2	667.3	82.8	9.0
Claude 2	2048	91.1	522.6	88.4	1.3
Claude 2	88000	90.3	551.5	87.1	2.0
Mixtral-8x7B	2048	81.5	679.1	85.9	4.1
LLaMA2-7B-Inst	2048	72.4	684.9	76.4	36.1
<i>Summaries generated via incremental updating</i>					
GPT-4	2048	82.5	805.4	84.1	3.4
GPT-3.5-Turbo	2048	67.0	484.5	68.2	3.5
Claude 2	2048	78.6	657.1	89.4	1.9
Claude 2	88000	90.9	493.7	84.7	1.9
Mixtral-8x7B	2048	64.5	558.7	82.3	3.5

GPT-4 summaries generated via incremental updating and hierarchical merging, respectively, while using LLM annotations yields a BOOOOKSCORE of **82.4** and **90.8**. Figure 4 compares the error distributions from GPT-4 to those of human annotators and shows that GPT-4 is more sensitive to omission errors and less sensitive to duplication or language errors. Taken as a whole, these results confirm that BOOOOKSCORE is a reliable annotator of coherence for book-length summarization. While we implement BOOOOKSCORE with GPT-4 for the remainder of this paper, implementing BOOOOKSCORE with open-source LLM annotators is an exciting future direction.

5 SYSTEMATIC EVALUATION OF LLMs

Armed with BOOOOKSCORE, we now investigate the impact of several critical implementation decisions on summary coherence, including the choice of prompting strategy, base LLM, and chunk size. Overall, Claude 2 produces the most coherent summaries as measured by BOOOOKSCORE, followed closely by GPT-4 and distantly by GPT-3.5-Turbo, Mixtral-8x7B, and LLaMA2-7B-Inst; however, GPT-4’s summaries are significantly longer and more detailed than the others across both prompting strategies. The rest of this section drills down into finer-grained results.

Experimental setup: Table 2 contains results for five instruction-tuned LLMs: GPT-4, GPT-3.5-Turbo, Claude 2, Mixtral-8x7B, and LLaMA2-7B-Instruct.¹³ Unless otherwise specified, we set the chunk size to 2048, maximum summary length G_n to 900, decoding temperature to 0.5,¹⁴ and $p = 1$ for ancestral sampling.¹⁵ To avoid confounds, we use identical prompts for all models except LLaMA2-7B-Inst, which only functions with a simpler prompt. LLM API costs for our experiments were \$10K USD (Table 8); more experimental details are in Appendix D.

Incremental summaries are almost always less coherent than their hierarchical counterparts. Hierarchical summaries generally have higher BOOOOKSCORE than incremental summaries, likely because the incremental updating task requires the base LLMs to follow more complex instructions

¹³GPT-4 configurations in this table are not comparable to the ones we analyzed in Section 3 since we had to reduce chunk size and summary length due to LLaMA2-7B-Inst and GPT-3.5-Turbo’s smaller context size.

¹⁴Claude 2 is the only exception, as we use its default temperature of 1.

¹⁵We use a temperature of 1 for compression, which improves adherence to the max summary length.

(e.g., deciding what to include from the current book chunk, what to discard from the summary, whether to restructure the summary, etc.). While hierarchical summarization potentially drops long-range dependencies, its instructions are generally simpler (summarize or merge).

Incremental summarization benefits from increased chunk size. The one exception to the above result is Claude 2 with a chunk size of 88K, whose incremental configuration produces slightly more coherent summaries than the hierarchical version (90.9 vs. 90.3 BOOOOKSCORE). In contrast, using Claude 2 for incremental summarization with a chunk size of 2048 results in a BOOOOKSCORE of 78.6, so clearly the model benefits from fewer updating and compression steps. We do not observe similar behavior with hierarchical summaries, which suggests that hierarchical book-length summarization is preferred for smaller context models.

LLaMA 2 struggles on book-length summarization while Mixtral shows promising performance. Table 2 shows that LLaMA-2-7B-Instruct achieves by far the worst hierarchical BOOOOKSCORE of any model. Its summaries also contain significant repetition (*% of repeated trigrams*), which is a critical coherence error. Furthermore, we could not get the LLaMA-2-7B-Instruct checkpoint to perform incremental updating at all, as it just copied text from the chunks until it reached the summary length limit, at which point it failed to follow the compression instruction. On the positive side, Mixtral-8x7B, another open-source LLM, outperforms LLaMA-2-7B-Instruct by a substantial margin, though it still trails behind most of the closed-source models. Nonetheless, it is encouraging to note that with performances closely matching that of GPT-3.5-Turbo on both summarization approaches, Mixtral-8x7B signals the narrowing gap between open-source and closed-source models.

High coherence does not necessarily correlate with human preferences. How well do coherence measurements from BOOOOKSCORE correlate with coarse-grained human preferences? We conduct another human evaluation study with the same four annotators in which we solicit preference judgments on pairs of GPT-4 generated incremental and hierarchical summaries.¹⁶ As shown in Table 4, incremental summaries are almost always preferred over hierarchical summaries in terms of level of detail (83% vs. 11%). However, hierarchical summaries are preferred for better structure (59% vs. 35%), logical consistency (53% vs 38%), and overall (54% vs. 44%). When forming their overall preference, some annotators preferred the higher level of detail of incremental summaries at the expense of coherence; thus, both strategies can be viable depending on the needs of the user.

Qualitative analysis: Appendix L contains summaries generated from Janika Oz’s *A History of Burning*, which tells a multi-generational story about an Indian family living in Uganda. We observe that both GPT-4 and GPT-3.5-Turbo tend to generate oft-repetitive and vague sentences within their summaries (e.g., *The story highlights the resilience and determination of the characters as they navigate the complexities of life, love, and identity across generations and continents.*). Such artifacts are rarely produced by the 88K chunk size version of Claude 2, which instead omits key information present in the beginning or middle of the input (e.g., the entire story of the first generation in the book) in favor of focusing on the end of the book, following the findings of Liu et al. (2023a). All configurations make faithfulness errors: for example, in *A History of Burning*, the mother of the character Hari is incorrectly identified as Rajni by Claude 2, while GPT-4 does describe Hari’s parentage correctly at one point in the summary but incorrectly at another. We show in Appendix I that automatic quality metrics such as BLANC (Vasilyev et al., 2020) and SUPERT (Gao et al., 2020) are inadequate for book-length summarization.

6 LIMITATIONS

Our error taxonomy is derived just from errors made by GPT-4. We decided to conduct our human evaluations in Section 3 on summaries produced by GPT-4 for two reasons: (1) we wanted our error taxonomy to focus on errors that are actually made by state-of-the-art LLMs (unlike e.g., fluency errors present in SNaC); and (2) human evaluation is very costly, so we could not evaluate many different LLMs on our annotation budget. Similarly, we implement BOOOOKSCORE using GPT-4 as a base LLM, which may have some systematic biases that could be alleviated by using a pool of LLM annotators as in AlpacaEval (Dubois et al., 2023).

¹⁶Each annotator compared 25 disjoint pairs of summaries, and we paid \$15 per task for a total of \$1.5K. To prevent bias, we shuffle the ordering of incremental and hierarchical summaries for each summary pair, and conceal the summarization method of each summary.

BOOOOKSCORE can be expensive to run. Since computing BOOOOKSCORE requires iterating through a summary sentence by sentence using GPT-4, it can be expensive and slow especially given that the annotation prompt is long (see Appendix M.4). We did experiment with an approach that asked GPT-4 to annotate errors in the entire summary at once, but the generated annotations would often include too many trivial questions, and alignment with human judgments was low. That said, despite the API costs of GPT-4 and the relatively slow time to evaluate one summary, BOOOOKSCORE is still significant cheaper and faster than performing human evaluations.

BOOOOKSCORE does not account for the relative importance of different error types. Unlike similar evaluation frameworks such as MQM (Freitag et al., 2021), we choose not to assign severity weights to different error types. Nowadays, powerful LLMs rarely make errors related to grammar, which can be objectively defined. For other error types like those in our taxonomy, the notion of assigning relative importance is ill-defined. Furthermore, prior work (Goyal et al., 2022a; Dou et al., 2022) shows low recall between human annotations for NLG evaluation, which indicates that error type severity is subjective as annotators often do not highlight issues that others may find critical.

No validation of recall. Due to the expense, we do not collect overlapping annotations for each summary during human evaluation. Since the annotation task involves subjectivity, overlapping annotations can help ensure that all errors within a summary can be captured. However, recent work (Krishna et al., 2023) shows that a comprehensive annotation of all information units is not required to produce a useful aggregate score that can be used to rank different models.

7 RELATED WORK

Book-length narrative summarization: Most prior long-form summarization work still focuses on documents shorter than 10K tokens (Cohan et al., 2018; Kornilova & Eidelman, 2019; Wang et al., 2022). BookSum (Kryscinski et al., 2022) is the first published summarization dataset that includes book-level source text as part of their data, which encouraged modeling efforts in this direction (Wu et al., 2021; Xiong et al., 2022; Pang et al., 2023; Cao & Wang, 2023; Pu et al., 2023a).

Fine-grained evaluation of generated text: Our work relates to evaluation protocols within machine translation that annotate spans, error types, and error severities (Freitag et al., 2021; Fernandes et al., 2023), which are more meaningful than output ranking and Likert ratings. Also related is ACU (Liu et al., 2023c), an annotation protocol for summary salience evaluation that breaks summaries down into fine-grained content units, FactScore (Min et al., 2023), which dissects machine-generated text into atomic facts before evaluating their factual consistency, LongEval (Krishna et al., 2023), which includes an in-depth analysis of best practices for faithfulness evaluation in long-form summarization coherence evaluation, and SNaC (Goyal et al., 2022a), a coherence error taxonomy built for fine-tuned summarization models.

Automatic evaluation with LLMs: LLM evaluators have recently emerged as a cost-effective alternative to human evaluations, explored for both general conversational and instruction following capabilities (Dubois et al., 2023; Zheng et al., 2023) and traditional NLG tasks like summarization (Fu et al., 2023; Liu et al., 2023b; Wang et al., 2023). These latter studies substantiate LLMs’ potential as an NLG metric, but only for evaluating short input-output pairs. In our work, we use GPT-4 to evaluate book-length summaries, uniquely employing a fine-grained automatic evaluation schema to set our work apart from existing research.

8 CONCLUSION

Our work presents the first systematic study of book-length summarization using LLMs. We establish a novel human evaluation protocol to assess summary coherence on newly-published books. Then, we develop an LLM-based automatic metric called BOOOOKSCORE that relies on a coherence error taxonomy derived from our human annotations. Using BOOOOKSCORE allows us to evaluate various prompting strategies and model choices, revealing insights such as: hierarchical merging produces more coherent summaries but may lack detail compared to incremental updating; and increasing chunk size can significantly improve incremental updating. Interesting future directions include automatically evaluating faithfulness in the book-length summarization setting, benchmarking newer long-context LLMs using BOOOOKSCORE, and expanding BOOOOKSCORE to multilingual texts. We release our BOOOOKSCORE metric and annotated summaries to enable meaningful progress in book-length summarization.